



Document Management System



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Title: PICU & High Dependency Admission & Discharge Criteria

1) Criteria for admission to PICU

• General:

-Age:below 13-year-old

-Weight: equalor more than 2 kg

-NoDNR patients unless temporary reversed for a medical reason orsevere DKA

-Anyadmission must be proved by pediatric intensivist on service/duty

I: Respiratory system

- Patients withsevere or potentially life-threatening pulmonary or airway disease.

Conditionsinclude, but are not limited to:

1. Endotrachealintubation or potential for emergency

endotrachealintubation and mechanical ventilation, regardless of etiology

2. Rapidlyprogressive pulmonary disease of high severity with risk of progressionto respiratory failure and/or total obstruction

despite trial of Non Invasive Ventilation (HFNC, CPAP, BIBAP) if applicable.

4. Newly placed tracheostomy with or without the need for mechanical ventilation

5. Patients who are not responding well or deteriorating despite multiple or high bronchodilator infusions or continuous inhaled or nebulized medications

6. Patients on non-invasive ventilation in High Dependency unit with increasing ventilator and Oxygen requirement.

II: Cardiovascular system

- Patients with severe, life-threatening, or unstable cardiovascular disease. Conditions include, but are not limited to:

1. Severe shock requiring more than one inotropic support or/and need for invasive monitoring.

2. Post-cardiopulmonary resuscitation

3. Life-threatening dysrhythmias

4. Unstable congestive heart failure, with or without mechanical ventilation

5. Congenital heart disease with unstable cardiorespiratory status

6. Post simple cardiovascular procedures with no cardiopulmonary bypass. ex: PDA ligation and PA banding.

7. Monitoring of arterial or central venous pressure.

9. Hypertensive urgency or emergency requiring intensive and invasive monitoring and management with anti-hypertensive infusions

III: Neurology

- Patients with actual or potential life-threatening or unstable neurologic disease.

Conditions include, but are not limited to:

1. Seizures, unresponsive to therapy or requiring high dose continuous infusion of anticonvulsive agents

coma with potential for airway compromise

3. Acute inflammation or infections of the spinal cord, meninges, or brain with neurologic depression, metabolic and hormonal abnormalities, respiratory or hemodynamic compromise or potential for increased intracranial pressure

4. Preoperative neurosurgical conditions with neurologic deterioration

5. Progressive neuromuscular dysfunction with or without altered sensorium requiring cardiovascular monitoring and/or respiratory support

IV: Hematology/oncology

- Patients with life-threatening or unstable hematologic or oncologic disease or active life-threatening bleeding.

Conditions include, but are not limited to:

1. Plasmapheresis or leukopheresis with unstable clinical condition

2. Anemia resulting in or with potential for hemodynamic and/or respiratory compromise

3. Complications of sickle cell crisis, such as neurologic changes, acute chest syndrome, or aplastic anemia with hemodynamic or respiratory instability

4. Initiation of chemotherapy with anticipated tumor lysis syndrome

5. Tumors or masses compressing or threatening to compress vital vessels, organs, or airway.

V: Endocrine/Metabolic

- Patients with life-threatening or unstable endocrine or metabolic disease. Conditions include, but are not limited to:

1. Severe diabetic ketoacidosis requiring frequent observation and therapy (PH < 7.0 and bicarb < 5)

2. Other electrolyte abnormalities, such as:

a. Persistent hyperkalemia requiring frequent monitoring and more intensive intervention or dialysis.

sensorium

- c. Hypo- or hyperglycemia requiring intensive monitoring
 - d. Severe metabolic acidosis requiring bicarbonate infusion, intensive monitoring, or complex intervention
 - e. Complex intervention to maintain fluid balance
3. Inborn disorders of metabolism with acute deterioration requiring respiratory support, acute dialysis, hemoperfusion, management of intracranial hypertension, or inotropic support.

VI: Gastrointestinal

-Patients with life-threatening or unstable gastrointestinal disease.

Conditions include, but are not limited to:

- 1. Acute gastrointestinal bleeding leading to hemodynamic or respiratory instability
- 2. Emergency endoscopy for removal of foreign bodies and requirement of intensive monitoring.
- 3. Acute hepatic failure leading to coma, hemodynamic or respiratory instability.

VII: Surgical

-Postoperative patients requiring frequent monitoring and potentially requiring intensive intervention.

Conditions include, but are not limited to:

- 1. Cardiovascular surgery who did not undergo cardiopulmonary bypass and do not have post-operative open sternum.
- 2. Thoracic surgery
- 3. General surgery with hemodynamic or respiratory instability or requirement of post-operative ventilation or invasive monitoring.
- 4. Organ transplantation other than heart or lung.

VIII: Renal system

-Patients with life-threatening or unstable renal disease.

Conditions include, but are not limited to:

- 1. Acute hemodialysis, peritoneal dialysis, or other continuous renal

2. Acute rhabdomyolysis with renal insufficiency.

IX: Multi-system and others

- Patients with life-threatening or unstable multi-system disease.

Conditions include, but are not limited to:

1. Toxic ingestions and drug overdose with potential acute decompensation of major organ systems
2. Multiple organ dysfunction
3. Suspected or documented malignant hyperthermia
4. Household or environmental (e.g., electrical or lightning) injuries

XI: Special intensive technological needs

- Conditions that necessitate the application of special technological needs, monitoring, complex intervention, or special treatment.

Discharge/Transfer Criteria

- Patients in the PICU will be evaluated and considered for discharge based on the reversal of the disease process or resolution of physiologic condition that prompted admission to the unit.

- Decision of discharge from PICU must be taken or confirmed with pediatric intensivist on service/duty.

- According to patient condition and requirement of monitoring and intervention, can be discharged to a designated patient care unit either high dependency units or wards.

- Transfer/discharge will be based on the following criteria:

1. Stable hemodynamic parameters. Arterial line should be removed if present.
2. Stable respiratory status and airway patency.
3. Continuous intravenous inotropic support, vasodilators, and antiarrhythmic drugs are no longer required or, when applicable, low doses of these medications can be administered safely in otherwise stable patients in a designated patient care unit
4. Controlled cardiac dysrhythmias
5. Neurologic stability with seizure control

has been reversed or resolved and are stable, may be discharged to a designated patient care unit that routinely manages chronically ventilated patients, when applicable, or to home

7. Routine peritoneal or hemodialysis with resolution of critical illness not exceeding general patient care unit guidelines

8. Patients with mature and stable artificial airways (tracheostomies) not requiring frequent suctioning of more than every 30 minutes.

2) Admission Criteria High Dependency Unit

• General:

-Age: below 13-year-old

-Weight: equal or more than 2 kg

-No DNR patients unless temporary reversed for a medical reason or DK requiring insulin infusion

I: RESPIRATORY SYSTEM

-Patients with moderate and non-life-threatening pulmonary or airway disease.

Conditions include, but are not limited to:

1. Requirement of minimal ventilator support via mature and stable tracheostomy.
2. Patients who are on any type of home nocturnal or full day ventilation only if they need higher ventilator settings or oxygen than their baseline but not sick enough to be admitted in PICU.
3. Rapidly progressive pulmonary disease of moderate severity with risk of progression to respiratory failure and/or total obstruction
4. Requirement of supplemental oxygen ($\text{FIO}_2 > 0.5$), regardless of etiology
5. Patients with mature tracheostomy requiring frequent suctioning
6. Patients requiring frequent ($< Q2H$), or continuous nebulized medications or bronchodilator infusions.
7. Patients requiring apnea work-up and cardiorespiratory

8. Patients requiring noninvasive respiratory support including

HFNC, CPAP or BiPAP

II: CARDIOVASCULAR SYSTEM

-Patients with moderate and non-life-threatening.

Conditions include, but are not limited to:

1. Patients with non-life-threatening cardiac disease requiring low dose inotropic or vasodilator therapy

2. Post-cardiac cath patients who require close monitoring with no respiratory or hemodynamic compromise

3. Non-life-threatening dysrhythmias with or without need for Cardioversion.

4. Patient with heart failure requiring close monitoring or noninvasive support.

III: NEUROLOGIC

-Patients with non-life-threatening neurologic disease.

Conditions include, but are not limited to:

1. Seizures, responsive to therapy but requiring continuous cardiorespiratory monitoring, those with no hemodynamic compromise but have the potential for respiratory compromise

2. Patient with altered sensorium in whom neurologic deterioration or depression is unlikely but require frequent neurologic assessment

3. Acute inflammation or infections of the spinal cord, meninges, or brain with noneurologic deficiency or other complications

4. Progressive neuromuscular dysfunction without altered sensorium requiring cardiorespiratory monitoring.

IV: HEMATOLOGY/ONCOLOGY

-Patients with non-life-threatening and potentially unstable hematologic

Conditions include, but are not limited to:

1. Plasmapheresis or leukopheresis with stable clinical condition
2. Severe anemia without hemodynamic and/or respiratory compromise
3. Moderate complications of sickle cell crisis, such as respiratory distress without acute chest syndrome or hemodynamic instability

V: ENDOCRINE/METABOLIC

-Patients with non-life-threatening and potentially unstable endocrine or metabolic disease.

Conditions include, but are not limited to:

1. Moderate diabetic ketoacidosis requiring frequent observation and therapy, without altered sensorium.
“DNR patients with DKA requiring insulin infusion are eligible for HD admission)
2. Other electrolyte abnormalities, such as:
 - a. Hyperkalemia, requiring cardiac monitoring and acute intervention
 - b. Severe hypo- or hyponatremia with seizures or altered sensorium
 - c. Hypo- or hyperglycemia requiring intensive monitoring
 - d. Hypo- or hypercalcemia requiring calcium infusion
 - e. Moderate metabolic acidosis requiring bicarbonate infusion
3. Inborn disorders of metabolism with acute deterioration requiring cardiorespiratory monitoring.

VI: GASTROINTESTINAL

-Patients with non-life-threatening or unstable gastrointestinal disease.

Conditions include, but are not limited to:

1. Acute gastrointestinal bleeding without hemodynamic or respiratory compromise

cardiorespiratory compromise

3. Chronic gastrointestinal or hepatobiliary insufficiency without coma, hemodynamic or respiratory instability.

VII: RENAL SYSTEM

-Patients with non-life-threatening and potentially unstable renal disease. Conditions include, but are not limited to:

1. Hypertension without seizures or encephalopathy who require close monitoring and frequent intermittent intravenous or oral medications
2. Non-complicated nephrotic syndrome with chronic hypertension requiring frequent blood pressure monitoring.

IX: MULTI-SYSTEM AND OTHER

-Patients with non-life-threatening and potentially unstable multi-system disease.

Conditions include, but are not limited to:

1. Uncomplicated toxic ingestions and drug overdose with no cardiorespiratory compromise, who require close monitoring
2. Post Pleural drain after initial stabilization
3. No patients should be admitted routinely as a temporary recovery from a procedure such as MRI, Line insertion, Bronchoscopy etc. unless has respiratory or hemodynamic issue or meeting any of other admission criteria.

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